Eliminating sunlight backflow in AR-HUDs through a Faraday rotator under the étendue constraint

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Abstract

This paper proposes an AR-HUD architecture incorporating a Faraday rotator to eliminate sunlight backflow, achieving a sunlight isolation of 2.39% across the visible spectrum. Furthermore, a time-multiplexing scheme doubles the effective étendue, enabling two virtual image distances while fully utilizing the étendue of the aperture-limited Faraday rotator.

Author Keywords

Head-up display; sunlight backflow; Faraday rotator; étendue.

1. Introduction

Head-up displays (HUDs) serve as in-vehicle human-machine interfaces by projecting driving information into a driver's eyes. Thus, HUDs allow drivers to access information without diverting their gaze downward, thereby significantly enhancing driving safety [1, 2]. A crucial reliability challenge stems from exposure to direct sunlight. The windshield, acting as a partial reflector with a higher transmittance than reflectance, allows sunlight to propagate backward along the imaging path. As shown in Fig. 1, the primary freeform mirror focuses incident sunlight onto the display panel, causing a rapid temperature rise that leads to permanent damage. Sunlight backflow detection and thermal management are compromising solutions that increase costs and system complexity [3, 4]. Furthermore, augmented reality (AR) HUDs feature a long virtual image distance (VID) and a wide field of view (FOV), causing the freeform mirror to collect sunlight over a wider angular range and focus the sunlight from infinity precisely onto the display panel [3, 5]. Consequently, the sunlight backflow issue is more severe in AR-HUDs.

To suppress sunlight backflow, commercial products primarily adopt specialized optical filters to remove the infrared spectrum or attenuate the P-polarized component of sunlight [3]. Those methods can reduce incident solar energy by approximately 50%, thereby protecting the display panel to some extent. Alternatively, projection-type display engines, such as digital light processing, laser beam scanning, and liquid crystal on silicon, can alleviate the backflow by employing a diffuser on the screen [5]. The diffuser's Lambertian properties scatter the concentrated sunlight, preventing it from being focused back onto the light engine. However, the micro-structured, non-planar diffuser introduces irregular scattering of the focused sunlight, degrading the contrast ratio under ambient illumination and potentially generating glare. Furthermore, projection-type engines incur higher costs and a bulky form factor.

Existing solutions only partially mitigate sunlight backflow, without fundamentally addressing the underlying degradation in display panel reliability, particularly in LCDs preferred for practical HUDs. To overcome this limitation, this study proposes using a Faraday rotator to block the concentrated sunlight propagating backward along the imaging path while maintaining display quality and efficiency.

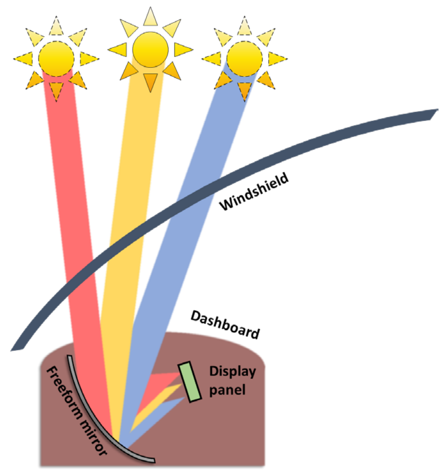

Fig. 1 Schematic optical path of sunlight backflow.

Recently, Ding et al. also incorporated a Faraday rotator into Pancake optics, enhancing the optical efficiency via a nonreciprocal path [6]. Despite the demonstrated potential of Faraday rotators for near-eye displays, HUDs have a much larger étendue than near-eye ones, namely a larger product of eyebox and FOV. The application of Faraday rotators in HUDs is constrained by the intrinsically small aperture of the magneto-optical crystal—the core component of a Faraday rotator—and the uniformity of the magnetic field. To address this limitation, we developed a time-multiplexing scheme to effectively increase the Faraday rotator's étendue in addition to geometric and polarization optical design. In our experiment, a Faraday rotator module (40 mm by 20 mm transversely) achieved excellent sunlight isolation (transmittance: 2.39%) across the visible spectrum. Integrated it into an AR-HUD prototype, this system projects virtual images with satisfactory light efficiency (87.1%) at two VIDs: a far image of 6° by 3° at 7.5 m and a near one of 5° by 2.5° at 2 m.

2. Method

2.1 Principle of the Faraday Rotator

Fig. 2 illustrates the principle of the Faraday rotator [7]. The rotation angle θ is governed by Eq. (1), while its direction depends solely on the magnetic field's orientation, independent of the light propagation direction. This nonreciprocal behavior contrasts with conventional reciprocal optical elements used in displays. The nonreciprocity of the Faraday rotator creates an irreversible optical path, providing the fundamental mechanism for eliminating sunlight backflow.

$$\theta(\lambda) = \int v(\lambda)\mathbf{B}(\mathbf{l}) \cdot d\mathbf{l}, \tag{1}$$

where $v(\lambda)$ denotes the Verdet constant of the crystal material, characterizing its wavelength-dependent magneto-optical properties, and $\mathbf{B}(\mathbf{l})$ represents the magnetic flux density along the

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propagation direction, which varies with the placement **l** of the light-magnetic field interaction.

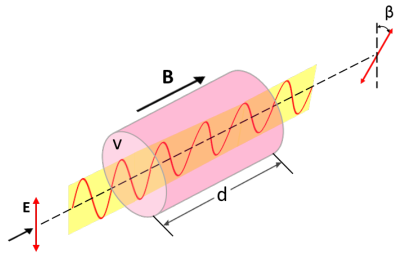

Fig. 2. Schematic of a Faraday rotator.

2.2 Proposed Design

Based on the Faraday rotator's principle, a 45° Faraday rotator is positioned between two linear polarizers whose transmission axes form a 45° angle. The operating principle of this optical isolator is depicted schematically in Fig. 3. In our study, the input polarizer's transmission axis is aligned at 45°, which matches the polarization direction of the light from the display panel. This configuration ensures minimal loss of display light. Conversely, sunlight propagating in reverse through the system is, in principle, completely blocked.

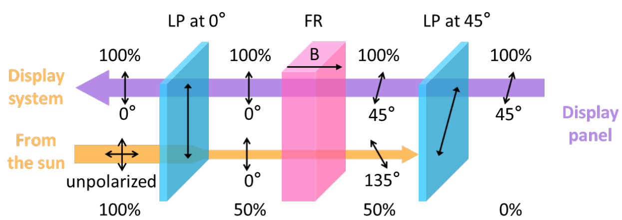

Fig. 3. The principle of the Faraday rotator-based optical isolator. Light propagates from the display panel to the display system and from the sun to the display panel. FR: Faraday rotator; LP: linear polarizer.

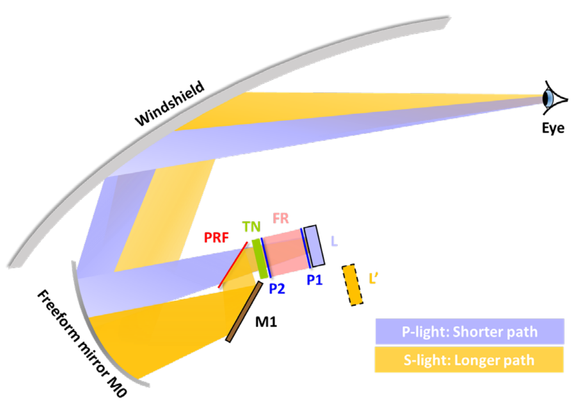

Fig. 4. Configuration of the proposed AR-HUD featuring an extended-étendue design and a Faraday rotator to eliminate sunlight backflow. PRF: polarizing reflective film; TN: TN LC; M1: plane mirror, P1: linear polarizer at 45°; P2: linear polarizer at 90°; L: display panel; L': equivalent display panel of S-polarized light.

However, the aperture size of the magneto-optical crystal and the magnetic field uniformity fundamentally constrain the system étendue, consequently limiting FOV and eyebox of the HUD. This limitation poses a significant challenge for AR-HUDs, which demand an extended FOV, enlarged eyebox, and far VIDs. To overcome this étendue constraint, we propose a time-multiplexing scheme [8]. This approach effectively doubles the usable étendue, enabling a dual-focal AR-HUD with excellent sunlight isolation.

Fig. 4 illustrates the proposed AR-HUD, leveraging the Faraday rotator principle to effectively suppress sunlight backflow while introducing negligible impact on the forward display path. A polarization-multiplexing unit, comprising a polarization converter using TN LC and a reflective polarizer, is positioned on the Faraday rotator's output side to face the display system. This configuration ensures compatibility between the polarization-dependent Faraday rotator and the polarization-multiplexing scheme, preventing polarization-state conflicts.

3. Experimental Verification

In the experiment, we chose TG28 as the magneto-optical crystal and NdFeB permanent magnets to provide sufficient, stable magnetic fields without ancillary cooling. Specifications of the proposed AR-HUD are summarized in Table 1. Based on the proposed configuration, an AR-HUD system was designed using Zemax OpticStudio. Its imaging quality was subsequently verified through optimization in LightTools.

Table 1. Specifications of the proposed AR-HUD.

	Far focal plane	Near focal plane
VID	7.5 m	2 m
FOV	6° by 3°	5° by 2.5°
Eyebox	70 mm by 30 mm	
Windshield	Windshield compatible with HUDs	
Eye-relief (driver to windshield)	800 mm	
Faraday rotator aperture	40 mm by 20 mm	

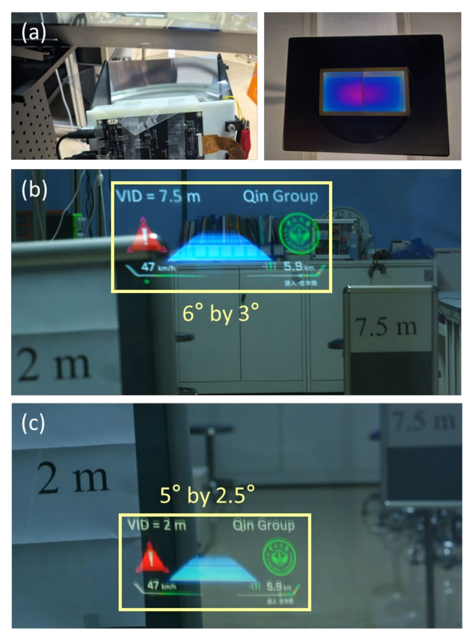

Fig. 5. (a) AR-HUD prototype with the proposed Faraday rotator and the actual isolation result. Captured virtual image at the (b) far (7.5 m) and (c) near (2 m) focal planes. FOVs are delineated by yellow rectangles.

Next, we built the AR-HUD system, assembled all components into a customized housing, and measured the optical isolation performance of the proposed Faraday rotator, as shown in Fig. 5(a). The freeform mirror spans 182 mm by 160 mm. A TFT-LCD that emits 45° polarized light serves as the picture generation unit. When driven by a ±24 V square wave at 50% duty cycle with zero bias, the TN LC cell switches the polarization state between P- and S-polarization, corresponding to the near and far virtual images, respectively. At switching frequencies above the critical flicker fusion threshold (~60 Hz), dual focal planes are merged.

For experimental validation, a camera (Fujifilm X-A7 with a 65 mm, f/2.8 lens) was positioned within the eyebox to capture virtual images. Virtual images at 7.5 m and 2 m are shown in Figs. 5(b) and (c), respectively. Real markers were placed at 7.5 m and 2 m in the background to demonstrate the different VIDs. The blurred one in either subfigure is out of the camera's focus.

Fig. 6 characterizes the key performance metrics of the proposed optical isolator: forward display transmission and backward sunlight rejection efficiency, the latter being the most critical parameter for suppressing backflow. The optical isolator exhibits a backward transmittance of 2.39% while maintaining a forward transmittance of 87.1%. At 485 nm, the transmittance decreases to 13.2% due to the magneto-optical crystal's intrinsic absorption, which corresponds to a region of low spectral intensity in the LCD output, thus minimizing the negative impact on the overall transmitted display spectrum. Consequently, the color performance is well preserved (see Fig. 5 for the high-fidelity colors). Combined with an infrared filter that can be simply added to an HUD product, this configuration offers decent sunlight backflow suppression.

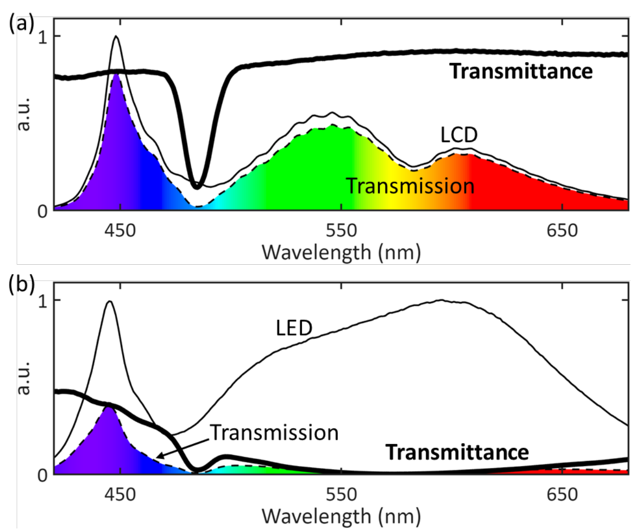

Fig. 6. Spectral response characterization of the proposed optical isolator: transmission spectrum and transmittance of the (a) forward path and (b) backward path. SP: spectrophotometer.

4. Conclusion

Sunlight backflow imposes a significant challenge on the display panel in AR-HUDs. This study presents a novel AR-HUD architecture incorporating a Faraday rotator to eliminate sunlight backflow. To overcome the constraint imposed by the Faraday rotator's limited aperture, we developed a time-multiplexing scheme that effectively doubles the étendue and enables two VIDs. Experimental results demonstrate effective optical isolation of 2.39% from sunlight backflow and satisfactory image quality at two focal planes: 6° by 3° at 7.5 m and 5° by 2.5° at 2 m, with a reasonable transmittance (87.1%)

5. References

[1] Liu Yi, et al. "A varifocal augmented reality head-up display using Alvarez freeform lenses," J. Soc. Inf. Display **32**(5), 226 (2024).

[2] Wei Ran, et al. "Large-size PVG-based waveguide simulation and uniformity optimization for AR-HUD," Opt. Express **32**(14), 25362 (2024).

[3] Wan Fushun, et al. "The challenge and development in AR-HUD," SID Symp. Dig. Tech. **55**(S1), 134 (2024).

[4] Xu Yang, et al. "Sun-focused spot detection method for AR-HUD," Proc. SPIE **13559**, 1699 (2025).

[5] Fan Chengxiang, et al. "Design of dual-focal-plane AR-HUD optical system based on a single picture generation unit and two freeform mirrors," Photonics **10**(11), 1192 (2023).

[6] Ding Yuqiang, et al. "Breaking the optical efficiency limit of virtual reality with a nonreciprocal polarization rotator," Opto-Electron. Adv. **7**(3), 230178 (2024).

[7] Carothers Kyle J., et al. "High Verdet constant materials for magneto-optical Faraday rotation: a review," Chem. Mater. **34**(6), 2531 (2022).

[8] Liu Yi, et al. "Compact dual-focal augmented reality head-up display using a single picture generation unit with polarization multiplexing," Opt. Express **31**(22), 35922 (2023).